

HDOC – Day 27

C Programming Assignment

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Batch 15

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Topic: Pyramid / Diamond Patterns using switch statement and nested loops (without arrays)

Problem Requirements

Display a menu for five patterns. Use a switch statement to select the pattern. Use nested loops without arrays. Each pattern is implemented in a separate parameterless function with no return value. For each activity, prepare the algorithm and test cases, then compile, execute, and test the C program.

Algorithm

Full pyramid

1. Read n.
2. For each row i from 1 to n: print (n-i) leading spaces, then print (2*i-1) asterisks separated by spaces.
3. Move to the next line.

Inverted pyramid

1. Read n.
2. For each row i from n down to 1: print (n-i) leading spaces, then print (2*i-1) asterisks separated by spaces.
3. Move to the next line.

Diamond pattern

1. Read n for the upper half.
2. Print the upper half using rows i = 1 to n.
3. Print the lower half using rows i = n-1 down to 1.
4. Each row has suitable leading spaces and (2*i-1) asterisks. Total rows =

$2*n-1$.

Palindromic number pyramid

1. Read n.
2. For each row i: print leading spaces.
3. Print numbers 1 through i in increasing order.
4. Then print i-1 through 1 in decreasing order.
5. Move to the next line.

Repeated-number pyramid

1. Read n.
2. For each row i: print leading spaces.
3. Print the row number i exactly ($2*i-1$) times.
4. Move to the next line.

C Program

```
#include <stdio.h>
void fullPyramid(void);
void invertedPyramid(void);
void diamondPattern(void);
void palindromicPyramid(void);
void repeatedNumberPyramid(void);
int main(void)
{
    int choice;
    printf("PYRAMID / DIAMOND PATTERN MENU\n");
    printf("1. Full pyramid\n");
    printf("2. Inverted pyramid\n");
    printf("3. Diamond pattern\n");
    printf("4. Palindromic number pyramid\n");
    printf("5. Repeated-number pyramid\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1: fullPyramid(); break;
```

```

case 2: invertedPyramid(); break;
case 3: diamondPattern(); break;
case 4: palindromicPyramid(); break;
case 5: repeatedNumberPyramid(); break;
default: printf("Invalid choice.\n");
}
return 0;
}

void fullPyramid(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) {
printf("Invalid number of rows.\n");
return;
}
for (i = 1; i <= n; i++)
{
for (j = 1; j <= n - i; j++)
printf(" ");
for (j = 1; j <= 2 * i - 1; j++)
printf("* ");
printf("\n");
}
}

void invertedPyramid(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) {
printf("Invalid number of rows.\n");
return;
}
}

```

```

for (i = n; i >= 1; i--)
{
for (j = 1; j <= n - i; j++)
printf(" ");
for (j = 1; j <= 2 * i - 1; j++)
printf("* ");
printf("\n");
}
}

void diamondPattern(void)
{
int n, i, j;
printf("Enter number of rows for upper half: ");
scanf("%d", &n);
if (n <= 0) {
printf("Invalid number of rows.\n");
return;
}
/* Upper half including middle row */
for (i = 1; i <= n; i++)
{
for (j = 1; j <= n - i; j++)
printf(" ");
for (j = 1; j <= 2 * i - 1; j++)
printf("* ");
printf("\n");
}
/* Lower half */
for (i = n - 1; i >= 1; i--)
{
for (j = 1; j <= n - i; j++)
printf(" ");
for (j = 1; j <= 2 * i - 1; j++)
printf("* ");
printf("\n");
}
}

```

```

}
}
void palindromicPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid number of rows.\n");
        return;
    }
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        for (j = i - 1; j >= 1; j--)
            printf("%d ", j);
        printf("\n");
    }
}

void repeatedNumberPyramid(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid number of rows.\n");
        return;
    }
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)

```

```

printf(" ");
for (j = 1; j <= 2 * i - 1; j++)
printf("%d ", i);
printf("\n");
}

}

```

Test Case Table

Test	Choice	n	Expected result
1	1	4	4-row full pyramid; rows contain 1, 3, 5, 7 asterisks
2	2	4	4-row inverted pyramid; rows contain 7, 5, 3, 1 asterisks
3	3	4	Diamond with 7 total rows; middle row has 7 asterisks
4	4	4	Palindromic rows ending with: 1 2 3 4 3 2 1
5	5	4	Repeated-number rows: 1; three 2s; five 3s; seven 4s
6	1	0	Invalid number of rows
7	9	-	Invalid choice.

Expected Outputs for n = 4

Choice 1 - Full pyramid

```

      *
    * * *
  * * * * *
* * * * * * *

```

Choice 2 - Inverted pyramid

```

* * * * * * *
 * * * * *
  * * *
   *

```

Choice 3 - Diamond

```

      *
    * * *
  * * * * *
* * * * * * *

```

```

* * * * *
* * *
*
```

Choice 4 - Palindromic number pyramid

```

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

Choice 5 - Repeated-number pyramid

```

1
2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
```

Compilation , Execution & Testing

```

Session Actions Edit View Help
(kali@kali)-[~]
$ vi hdoc27.c

(kali@kali)-[~]
$ cc hdoc27.c

(kali@kali)-[~]
$ ./a.out
PYRAMID / DIAMOND PATTERN MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 1
Enter number of rows: 4
*
* * *
* * * * *
* * * * * * *

(kali@kali)-[~]
$ ./a.out
PYRAMID / DIAMOND PATTERN MENU
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 2
Enter number of rows: 4
* * * * * * *
* * * * *
* * *
*
```

Session Actions Edit View Help

```
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 3
Enter number of rows for upper half: 4
```

```
  *
 * * *
* * * * *
* * * * *
 * * * *
  * * *
   *
```

(kali@kali)-[~]

\$./a.out

PYRAMID / DIAMOND PATTERN MENU

```
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 4
Enter number of rows: 4
```

```
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

(kali@kali)-[~]

\$./a.out

PYRAMID / DIAMOND PATTERN MENU

```
1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
Enter your choice: 5
Enter number of rows: 4
```

```
  1
 2 2 2
 3 3 3 3 3
4 4 4 4 4 4 4
```